

AI CAREER FOR WOMEN

Engineer Spoke



KOREAN FOOD RECIPE CLASSIFIER

A Project Report
submitted in partial fulfillment of the requirements
of
AICW project.

By

D.Rajitha,disiriporajitha@gmail.com
G.Likhitha Devi,likhidevi0530@gmail.com
G.Srimani,gundugara449@gmail.com
CH.sowjanya,chikkireddysowjanya@gmail.com

Under the Guidance of
Mr. Abdul Aziz Md(co-lead AICW)

ACKNOWLEDGEMENT

We would like to take this opportunity to express our deep sense of gratitude to all individuals who helped us directly or indirectly during this thesis work.

Firstly, we would like to thank my supervisor, **Mr. Abdul Aziz Md** for being a great mentor and the best adviser I could ever have. His advice, encouragement and the critics are a source of innovative ideas, inspiration and causes behind the successful completion of this project. The confidence shown in me by him was the biggest source of inspiration for me. It has been a privilege working with him for the last one year. He always helped me during my project and many other aspects related to the program. His talks and lessons not only help in project work and other activities of the program but also make me a good and responsible professional.

We are also profoundly grateful to the Head of the Department, esteemed faculty members, and technical staff for providing the necessary facilities, infrastructure, and a supportive academic environment that enabled us to carry out this research successfully.

Finally, we extend our sincere appreciation to our family members and friends for their unwavering encouragement, understanding, and moral support throughout the course of this endeavour.

The Korean Food Recipe Classifier is an artificial intelligence and machine learning-based project designed to identify and classify different types of Korean food from images and provide useful information about the corresponding recipes. Korean cuisine consists of a wide variety of traditional dishes such as Kimchi, Bibimbap, Bulgogi, Tteokbokki, Japchae, Kimbap, Samgyeopsal, and many others. Identifying these dishes manually can be difficult, especially for people who are unfamiliar with Korean cuisine. The proposed system aims to solve this problem by developing an automated food classification model that can recognize Korean dishes based on their visual features. The system uses a dataset containing images of different Korean food categories. The collected images are preprocessed through techniques such as resizing, normalization, and data augmentation to improve the quality and variety of the training data. Machine learning or deep learning algorithms are then used to train the classification model. During prediction, the user provides an image of a Korean dish, and the trained model analyzes the image and predicts the most likely food category. After classification, the system can display the name of the dish along with its recipe, ingredients, preparation method, and other relevant details. The main objective of the project is to make Korean food identification simple, fast, and accessible to users. It can be useful for students, tourists, food lovers, restaurants, and people interested in learning about Korean cuisine. The system also demonstrates the practical application of computer vision, image processing, and artificial intelligence in the food industry. The performance of the model can be evaluated using parameters such as accuracy, precision, recall, and F1-score. In the future, the system can be improved by increasing the size of the dataset, adding more Korean food categories, supporting real-time camera-based recognition, and providing nutritional information and multilingual recipe descriptions. Overall, the Korean Food Recipe Classifier provides an efficient and user-friendly approach for recognizing Korean dishes and accessing their recipe information using modern machine learning technologies.

TABLE OF CONTENTS

Abstract	
----------	-------

Chapter 1. Introduction

- 1.1 Problem Statement
- 1.2 Motivation
- 1.3 Objectives
- 1.4. Scope of the Project

Chapter 2. Literature Survey

Chapter 3. Proposed Methodology

Chapter 4. Implementation and Results

Chapter 5. Discussion and Conclusion

References

CHAPTER 1

Introduction

1. Background

Korean cuisine is popular around the world because of its unique flavors, ingredients, cooking methods, and traditional dishes. Foods such as Kimchi, Bibimbap, Bulgogi, Tteokbokki, Japchae, and Kimbap have become widely recognized. However, people who are unfamiliar with Korean cuisine may find it difficult to identify different dishes by simply looking at their appearance.

2. Need for the Project

With the growth of Artificial Intelligence (AI) and Machine Learning (ML), image-based food recognition has become an important application of computer vision. A **Korean Food Recipe Classifier** can automatically recognize a Korean food item from an uploaded image and provide its name and recipe details. This reduces the need for manual identification and makes information about Korean food easier to access.

3. Problem Statement

Identifying Korean food dishes manually can be challenging because many dishes have similar appearances and may contain unfamiliar ingredients. There is a need for an automated system that can accurately classify Korean food images and provide useful recipe information such as ingredients and preparation methods.

4. Proposed System

The proposed **Korean Food Recipe Classifier** uses image processing and machine learning techniques to classify different Korean food items. A dataset containing images of various Korean dishes is collected and preprocessed. The images are then used to train a classification model. When a user uploads an image, the trained model analyzes its visual features and predicts the corresponding Korean food category.

5. Objectives

The main objective of this project is to develop a simple and efficient system for recognizing Korean food using images. The system aims to improve food identification, provide recipe information, reduce manual effort.

CHAPTER 2

Literature Survey

Literature Survey – Korean Food Recipe Classifier

1. Introduction to Food Image Classification

Food image classification is an important application of **Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision**. Researchers have developed different techniques to identify food items from images based on features such as shape, color, texture, and appearance. These techniques help in automatic food recognition and can be applied to recipe recommendation, dietary analysis, restaurant applications, and food information systems.

2. Traditional Image Processing Techniques

Earlier food classification systems mainly used traditional image processing methods. Features such as **color, texture, shape, and edges** were extracted from food images. Machine learning algorithms such as **Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Trees, and Random Forest** were then used for classification. Although these methods can provide good results for simple datasets, their performance may decrease when images have different backgrounds, lighting conditions, or similar-looking food items.

3. Machine Learning-Based Food Classification

Machine learning has improved the ability of computer systems to recognize different food categories. Classification models are trained using labeled food-image datasets. During training, the model learns the visual characteristics of each food category and uses these learned patterns to classify new images. Such systems have been used for recognizing different cuisines and individual food items.

4. Deep Learning for Food Recognition

Deep learning, particularly **Convolutional Neural Networks (CNNs)**, has become widely used for image classification. CNN models automatically learn important visual features from images without requiring extensive manual feature extraction. Models such as **AlexNet, VGG, ResNet, Inception, and EfficientNet** have demonstrated strong performance in image recognition tasks. These approaches are suitable for Korean food classification because Korean dishes can have complex shapes, colors, textures, and presentations.

CHAPTER 3

Proposed Methodology

3. Proposed Methodology

3.1 System Design

The proposed **Korean Food Recipe Classifier** is designed to identify Korean food items from uploaded images and provide the corresponding recipe information. The system consists of different stages such as user input, image preprocessing, food image classification, recipe retrieval, and result display. The user first provides an image of a Korean food item. The image is then processed to improve its quality and prepare it for classification. The trained machine learning/deep learning model extracts important visual features from the image and predicts the food category. Based on the predicted food name, the system retrieves the related recipe details such as ingredients and preparation method and displays the results to the user.

3.1.1 Registration

In the proposed system, registration is used to create a user account and provide secure access to the application. The user enters basic details such as name, email ID, username, and password. These details are validated and stored in the system database. After successful registration, the user can log in and access the Korean food classification features. Registration helps the system maintain user-specific information and provides a simple and organized way to use the application.

3.1.2 Recognition

Recognition is the main component of the Korean Food Recipe Classifier. After logging into the system, the user uploads an image of a Korean food item. The uploaded image undergoes preprocessing such as resizing, normalization, and noise reduction. The processed image is given as input to the trained classification model. The model analyzes the visual characteristics of the food and predicts the most suitable Korean food category. After recognition, the system displays the predicted food name along with its recipe information, including ingredients, preparation steps, and other relevant details.

3.2 Modules Used

The proposed Korean Food Recipe Classifier can be divided into the following major modules:

3.2.1 Image Input Module

This module allows the user to upload or select an image of a Korean food item. The uploaded image acts as the input for the classification system. The system accepts commonly used image formats and forwards the image to the preprocessing stage.

3.2.2 Image Preprocessing Module

The preprocessing module prepares the input image for classification. The image is resized to the required dimensions and normalized to maintain consistent input values. Data augmentation techniques such as rotation, flipping, and zooming can also be used during model training to improve the model's ability to recognize food images under different conditions.

3.2.3 Food Classification Module

This is the core module of the system. A machine learning or deep learning model, such as a **Convolutional Neural Network (CNN)**, is trained using a dataset of Korean food images. The model learns important features such as color, shape, texture, and overall appearance. When a new image is provided, the trained model predicts the corresponding Korean food category.

3.2.4 Recipe Retrieval Module

After the food item is classified, this module retrieves the recipe associated with the predicted food name. It provides information such as the list of ingredients, preparation procedure, cooking method, and other relevant details. This makes the application more useful than a simple image classification system.

3.2.5 Result Display Module

The final prediction and recipe information are displayed to the user through the application interface. The result may include the food name, classification confidence, ingredients, and preparation instructions. The interface is designed to provide the information in a simple and understandable format.

3.2.6 Database Module

The database stores information related to users, food categories, images, and recipes. It allows the system to efficiently retrieve the recipe details after the food item has been identified.

3.3 Data Flow Diagram

The Data Flow Diagram (DFD) represents how information moves through the Korean Food Recipe Classifier. The user first registers or logs into the system and then uploads an image of a Korean food item. The image is sent to the preprocessing module, where it is resized and normalized. The processed image is then passed to the trained classification model. The model predicts the food category and sends the result to the recipe retrieval module. The recipe database is searched using the predicted food name, and the corresponding ingredients and preparation instructions are retrieved. Finally, the food name and recipe details are displayed to the user.

DFD Flow:

User → Registration/Login → Upload Food Image → Image Preprocessing → Classification Model → Food Prediction → Recipe Database → Recipe Retrieval → Result Display → User

This methodology provides an organized workflow for recognizing Korean food images and connecting the classification result with useful recipe information.

3.3.1 DFD Level 0

The **Level 0 Data Flow Diagram** represents the overall working of the Korean Food Recipe Classifier as a single process. The user interacts with the system by providing a Korean food image as input. The system processes the image and classifies it into the appropriate food category. The classified food name is then used to retrieve the corresponding recipe information from the recipe database. Finally, the system displays the food name, ingredients, and preparation method to the user.

Data

Flow:

User → Korean Food Recipe Classifier → Food Classification → Recipe Database → Recipe Details → User

3.3.2 DFD Level 1 – Food Image Input and Preprocessing Module

This module describes the detailed flow of the food image before classification. The user uploads an image of a Korean food item through the application. The system checks the input image and performs preprocessing operations such as resizing, normalization, noise removal, and image enhancement. The processed image is then forwarded to the classification model for further analysis.

Data

Flow:

User → Upload Food Image → Image Validation → Image Preprocessing → Processed Image → Classification Module

3.3.3 DFD Level 1 – Korean Food Recognition Module

The Korean Food Recognition Module is the main component of the system. The preprocessed food image is provided as input to the trained machine learning/deep learning model. The model extracts important visual features such as shape, color, texture, and patterns. These features are compared with the learned food categories, and the model predicts the most appropriate Korean food item. The predicted food name is then sent to the recipe retrieval module.

Data

Flow:

Processed Image → Feature Extraction → Trained Classification Model → Food Prediction → Food Name

3.3.4 DFD Level 1 – Recipe Retrieval Module

The Recipe Retrieval Module uses the predicted food name to search the recipe database. Once the food category is identified, the system retrieves the corresponding recipe information. The information may include the food name, ingredients, preparation steps, cooking method, and other relevant details. The retrieved information is then displayed to the user.

Data	Flow:
Food Prediction → Recipe Database → Recipe Search → Ingredients & Preparation Steps → Result Display	

3.4 Advantages

The Korean Food Recipe Classifier provides several advantages. It automatically identifies Korean food items from images and reduces the effort required for manual identification. The system provides quick and convenient results and helps users learn about different Korean dishes. It can classify multiple food categories when trained with a suitable dataset. The use of machine learning and image processing improves the efficiency of food recognition. The system can also provide recipe information along with the predicted food name, making it more useful for food lovers, students, tourists, and people interested in Korean cuisine. The system can be further expanded by adding more food categories and additional recipe information.

3.5 Requirement Specification

Requirement specification describes the hardware and software resources required to develop and operate the Korean Food Recipe Classifier. The requirements should be sufficient to support dataset processing, model training, image classification, database operations, and application execution.

3.5.1 Hardware Requirements

- **Processor:** Intel Core i3 or above
- **RAM:** Minimum 4 GB, preferably 8 GB or more
- **Storage:** Minimum 10 GB free space
- **Input Device:** Keyboard, mouse, and camera/image upload facility

- **Display:** Standard monitor with suitable screen resolution
- **GPU:** Optional, but an NVIDIA GPU can be used for faster deep learning model training
- **Internet Connection:** Required for downloading datasets, libraries, and other resources when needed

3.5.2 Software Requirements

- **Operating System:** Windows 10/11, Linux, or macOS
- **Programming Language:** Python
- **Development Environment:** Jupyter Notebook, Google Colab, or Visual Studio Code
- **Machine Learning Libraries:** TensorFlow/Keras, Scikit-learn
- **Image Processing:** OpenCV, PIL/Pillow
- **Data Processing:** NumPy and Pandas
- **Database:** MySQL, SQLite, or another suitable database
- **Frontend:** HTML, CSS, and JavaScript or a suitable Python-based interface
- **Backend:** Python with Flask/Django, if a web-based application is implemented
- **Web Browser:** Google Chrome, Microsoft Edge, or Mozilla Firefox

These requirements provide the necessary environment for developing, training, testing, and deploying the **Korean Food Recipe Classifier** efficiently.

CHAPTER 4

Implementation and Result

Implementation and Results

4.1 Implementation

The Korean Food Recipe Classifier is implemented using machine learning and image processing techniques to identify different Korean food items from images. The implementation consists of several stages, including dataset collection, image preprocessing, model training, food classification, and recipe retrieval. A dataset containing images of different Korean dishes such as Kimchi, Bibimbap, Bulgogi, Tteokbokki, Kimbap, and Japchae is used for developing the system.

4.1.1 Dataset Collection

A suitable dataset of Korean food images is collected for training and testing the classification model. Each image is assigned to its respective food category. The dataset is divided into training and testing sets so that the performance of the trained model can be evaluated on previously unseen images.

4.1.2 Image Preprocessing

The collected images may have different sizes, resolutions, and backgrounds. Therefore, preprocessing is performed before training the model. The images are resized to a fixed dimension and normalized to improve consistency. Data augmentation techniques such as rotation, flipping, zooming, and shifting can be applied to increase the variety of training images and reduce overfitting.

4.1.3 Model Training

A Convolutional Neural Network (CNN) or a suitable transfer-learning model is used for Korean food classification. During training, the model learns important features from the food images, including shape, color, texture, and visual patterns. The training process is performed over multiple epochs, and the model parameters are updated based on the classification error.

4.1.4 Food Recognition

After training, a new Korean food image is provided to the system. The image undergoes the same preprocessing steps used during training. The trained model analyzes the image and predicts the most probable food category. The predicted result is then displayed to the user.

4.2 Results

The implemented system successfully demonstrates the process of recognizing Korean food images and providing their related recipe information. When an image is uploaded, the system preprocesses the image and passes it to the trained classification model. The model predicts the corresponding Korean food category and displays the result along with the available recipe details.

The performance of the classifier can be evaluated using metrics such as accuracy, precision, recall, and F1-score. The classification accuracy depends on factors such as the size and quality of the dataset, number of food categories, image variations, and model architecture. A well-prepared dataset and appropriate preprocessing techniques can improve the recognition performance.

4.2.1 Overall Result

The Korean Food Recipe Classifier provides an automated and efficient method for identifying Korean food from images. The system reduces manual identification effort and provides recipe information immediately after classification. The project demonstrates the practical application of Artificial Intelligence, Computer Vision, Machine Learning, and Image Processing in the food domain. The system can be further improved by using a larger dataset, advanced deep learning architectures, real-time camera recognition, nutritional information, and support for additional Korean food categories.

CHAPTER 5

Discussion and Conclusion

5.1 Discussion

The Korean Food Recipe Classifier demonstrates how Artificial Intelligence, Machine Learning, and Computer Vision can be applied to identify Korean food items from images. The system takes a food image as input, preprocesses the image, and uses a trained classification model to predict the corresponding Korean food category. After recognizing the food, the system retrieves and displays the related recipe information, including ingredients and preparation methods.

The performance of the system depends mainly on the quality and size of the dataset, image preprocessing techniques, and the classification algorithm used. Images with different backgrounds, lighting conditions, food arrangements, and presentation styles can make classification more challenging. Using data augmentation and suitable deep learning techniques can improve the model's ability to recognize food under different conditions. The system also provides a convenient way for users to learn about Korean cuisine without requiring prior knowledge of individual dishes.

One of the important advantages of the proposed system is that it combines food recognition with recipe retrieval. Instead of only displaying the name of the recognized food, the system can provide useful recipe details to the user. This makes the project more practical for students, tourists, food enthusiasts, restaurants, and people interested in Korean cuisine. However, the system can still be improved by increasing the dataset, adding more food categories, and using more advanced classification models.

5.2 Conclusion

The Korean Food Recipe Classifier successfully provides an automated approach for identifying Korean food items using image classification techniques. The system processes the uploaded food image, extracts important visual features, classifies the image into the appropriate Korean food category, and provides the corresponding recipe information. The project reduces the effort involved in manually identifying Korean dishes and makes recipe information easily accessible.

The project demonstrates the effective use of Machine Learning, Deep Learning, Image Processing, and Computer Vision for solving a real-world food classification problem. With further improvements such as a larger and more diverse dataset, real-time camera-based recognition, nutritional information, multilingual support, ingredient detection, and improved deep learning models, the system can become more accurate and useful. Overall, the Korean Food Recipe Classifier is a practical and user-friendly application that combines food recognition and recipe information in a single system.

REFERENCES

REFERENCES

- Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet Classification with Deep Convolutional Neural Networks. *Advances in Neural Information Processing Systems (NeurIPS)*, 25.
- Simonyan, K., & Zisserman, A. (2014). Very Deep Convolutional Networks for Large-Scale Image Recognition. *International Conference on Learning Representations (ICLR)*.
- He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778.
- Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. *Proceedings of the International Conference on Machine Learning (ICML)*, 6105–6114.
- Chollet, F. (2017). Xception: Deep Learning with Depthwise Separable Convolutions. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 1251–1258.
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
- Szeliski, R. (2022). Computer Vision: Algorithms and Applications. Springer.
- TensorFlow Documentation. TensorFlow: An End-to-End Open Source Machine Learning Platform. [TensorFlow Official Documentation](#)
- Keras Documentation. Deep Learning for Humans. [Keras Official Documentation](#)
- OpenCV Documentation. Open Source Computer Vision Library. [OpenCV Official Documentation](#)
- Scikit-learn Documentation. Machine Learning in Python. [Scikit-learn Official Documentation](#)
- Python Documentation. Python Programming Language Documentation. [Python Official Documentation](#)

APPENDICES(if applicable)

Supplementary Materials – Korean Food Recipe Classifier

Appendix A – Sample Dataset Structure

The dataset used for the Korean Food Recipe Classifier can be organized into separate folders for each Korean food category. Each folder contains images belonging to that particular food class.

S.No Food Category Sample Images Purpose

1	Kimchi	Multiple images Training and classification
2	Bibimbap	Multiple images Training and classification
3	Bulgogi	Multiple images Training and classification
4	Tteokbokki	Multiple images Training and classification
5	Kimbap	Multiple images Training and classification
6	Japchae	Multiple images Training and classification

The dataset can be divided into training, validation, and testing datasets to train the model and evaluate its performance.

Appendix B – Sample Dataset Directory

Korean_Food_Classifier/

```
|
|  └─ dataset/
|  |  └─ train/
|  |  |  └─ kimchi/
|  |  |  └─ bibimbap/
|  |  |  └─ bulgogi/
|  |  |  └─ tteokbokki/
|  |  |  └─ kimbap/
|  |  |  └─ japchae/
|  |
|  |  └─ validation/
|  |  |  └─ kimchi/
|  |  |  └─ bibimbap/
|  |  |  └─ bulgogi/
|  |  |  └─ tteokbokki/
|  |
|  |  └─ test/
|  |  |  └─ kimchi/
```

```
|   ├── bibimbap/
|   ├── bulgogi/
|   └── tteokbokki/
|
|   ├── model/
|   └── korean_food_model.h5
|
└── application/
    └── app.py
```

Appendix C – Sample Image Preprocessing Code

The following Python code demonstrates the basic preprocessing of a food image before classification.

```
from PIL import Image
import numpy as np

image = Image.open("food_image.jpg")

# Resize image
image = image.resize((224, 224))

# Convert image into an array
image_array = np.array(image)

# Normalize pixel values
image_array = image_array / 255.0

# Add batch dimension
image_array = np.expand_dims(image_array, axis=0)

print("Image preprocessing completed.")
```

KOREAN FOOD RECIPE CLASSIFIER

Input Image: food_image.jpg

Predicted Food : Bibimbap

Confidence : 94.25%

Recipe:

Ingredients:

- Cooked rice
- Mixed vegetables
- Egg
- Korean sauce

Preparation:

1. Prepare the cooked rice.
2. Cook the vegetables separately.
3. Arrange the vegetables over the rice.
4. Add the egg and Korean sauce.
5. Mix well and serve.